Graph ML Model

Project 2

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Introduction

Material properties, such as drug receptor binding, are determined by atomic charges. Quantum mechanical has been used for predicting atomic charges. However, this mechanism requires a significant allocation of computational resources, making it an inefficient approach for scaling. Graph Neural Networks (GNNs) are designed to operate on a graph-structured data consisting of nodes and edges. Each of these nodes contain a feature vector, and they are constantly being compiled with new information from other neighboring nodes through message passing. Through the multiple rounds of message passing, each of these nodes gain the ability to make predictions at either the node, edge, or graph level, making it the prime ML choice to perform atomic charge prediction tasks. Compared to the quantum mechanical methods, GNN can make these predictions at a significantly lower computational cost while maintaining its accuracy. This has made GNNs the prime choice for predicting atomic charges from the crystal structures in POSCAR format directly.

In this graph ML model, we transformed the raw POSCAR and atomic charge files into a PyTorch Geometric graph dataset. Each crystal structure is graphically converted with atoms represented as nodes and the interatomic interactions represented as edges. A two-stage hyperparameter search was conducted across 25 total configurations to identify the best performing model, which included exploring combinations of hidden layers, total number of neuron layers, dropout rates, activation functions, optimizers, learning rate, and batch size. The configurations with the most optimized output from the first stage were moved forward into the second refining stage, where their validation loss was assessed. The final refining stage consisted of prediction accuracy check and residual diagnostics to determine the strength and weakness of these layers in predicting atomic charges from their crystal structures.

Dataset Description

The POSCAR files were used on top of the atomic charge files to define the atomic structures and the corresponding charge for each atom in the structure. The descriptive title for each of the files in POSCAR is embedded in a comment line, with the 3x3 matrix of lattice vectors defining the dimensions of the unit cells formatted as a plain text file document. The specific details of each element, including its symbol, atom counts and their positions are listed in different rows separated by the different elements. The complementary atomic charge documentation provides one charge value per atom in the same order as the atomic positions in the POSCAR file. From the scatterplot of the prediction, the approximated charge value in this dataset ranges from -2.5 to +2.5. There are two distinctive clusters defined in this scatterplot, with their centroids located around -1.2 and +1.3. The distance between these two centroids suggests that the element types within this dataset have opposing charge characteristics.

The POSCAR file uses Periodic Boundary Condition (PBC) to simulate infinite crystalline structures using a finite unit cell that repeats infinitely in all directions. In this environment, atoms near one boundary interact with atoms near the opposite boundary, eliminating artificial surface effect and ensuring physically realistic simulations. The `get_all_distances (mic=True)` function within the Atomic Simulation Environment (ASE) library was used to properly account for PBC and calculates pairwise distance between all atoms, all while applying the Minimum Image Convention to handle periodic interactions accurately.

Methodology

To allow this GNN model to learn directly from the structural relationships between each atom in a crystal structure, the POSCAR and atomic charge file need to be converted into a graph dataset that is compatible with PyTorch Geometric. The entire crystal structure was represented as a graph, where the atoms serve as different nodes across the graph. Edges were established as connections between two atoms when they are within a certain distance threshold of each other. Atom specific properties, including atomic number and one-hot element encoding, were used to construct node features, allowing the model to distinguish between different element types. The ASE's `get_all_distances(mic=True)` function was used to compute the PBC-aware pairwise distance, an important factor when determining edge connection; ensuring that interactions across periodic boundaries were property captured. Interatomic distances, encoded as edge attributes, provide the model with geometric context about the surrounding atomic environment. Each of these structures was then stored as a PyTorch Geometric data object, which contains the node's feature matrix, edge index, edge attributes, and the corresponding atomic charge value as target, completing the graph transformation needed for GNN training.

The GNN model was designed to predict per-atom charge through a series of graph convolutional layers, aggregating feature information from neighboring nodes through multiple rounds of message passing. During this process, the model can build a richer representation of each atom's local chemical environment across successive layers, therefore improving the accuracy of its prediction. Through multiple rounds of hyperparameter tuning, the best performing model configuration consisted of three convolutional layers with 64 neurons per layer with the ELU activation function and a 10% dropout rate. The output from the final convolutional layer was passed through another fully connected layer, mapping each node's embedding to a single scalar value that represents the predicted atomic charge.

A two-stage hyperparameter search across 25 different configurations was assessed to identify the most optimal configuration for the model. The first stage consisted of 20 individual trials that explored the hyperparameter space, where each trial assessed a different combination of architectural and training choices. This includes the number of hidden layers, optimizer types, ideal learning rate, and regularization settings. The results from the first stage shows a wide range in validation loss from approximately 0.13 to 5.9, highlighting the sensitivity of how hyperparameter choice can directly impact the model's performance. The top five configurations chosen from stage 1 were further evaluated in stage 2, where they were retrained more carefully based on their validation loss. From the two-stage search, AdamW optimizer with a fine-tuned learning rate of 0.0005 and a weight decay of 0.001 trained in a small batch size of 2 achieves the best validation loss of 0.09.

The training and evaluation process for this GNN model resembles the process of evaluating other ML models, with the entire dataset being split into three parts – training, validation, and testing. Each of these processes uses a portion of the dataset, ensuring that the model is making predictions based on past training patterns and not individual samples. Since atomic charges experience continuous regression, Mean Square Error (MSE) loss was used as the training objective. Dropout regularization and weight decay techniques were implemented to prevent overfitting throughout the training process using AdamW, and early stopping was also included to stop the training at the best model checkpoint with the lowest validation loss. All these regularization strategies ensured the model is generalizing the learned data plots instead of memorizing samples of training data.

Results

A combination of training dynamics, prediction accuracy and residual analysis was used to evaluate this GNN model. Through the analysis of the training and validation loss curves recorded over 100 epochs, both the training and validation losses have both experienced a steady and consistent drop. This consistent decrement in these loss curves signifies that the model is progressively learning the fundamental structural patterns of atomic charges from crystal structure. As training progressed, both curves continued to decrease at a slower pace and have converged closer to each other, with the best validation loss achieved at Epoch 87. The small variations between the two loss curves throughout this entire training process further confirms the model is generalizing well to unknown data without overfitting the model with sample-specific features.

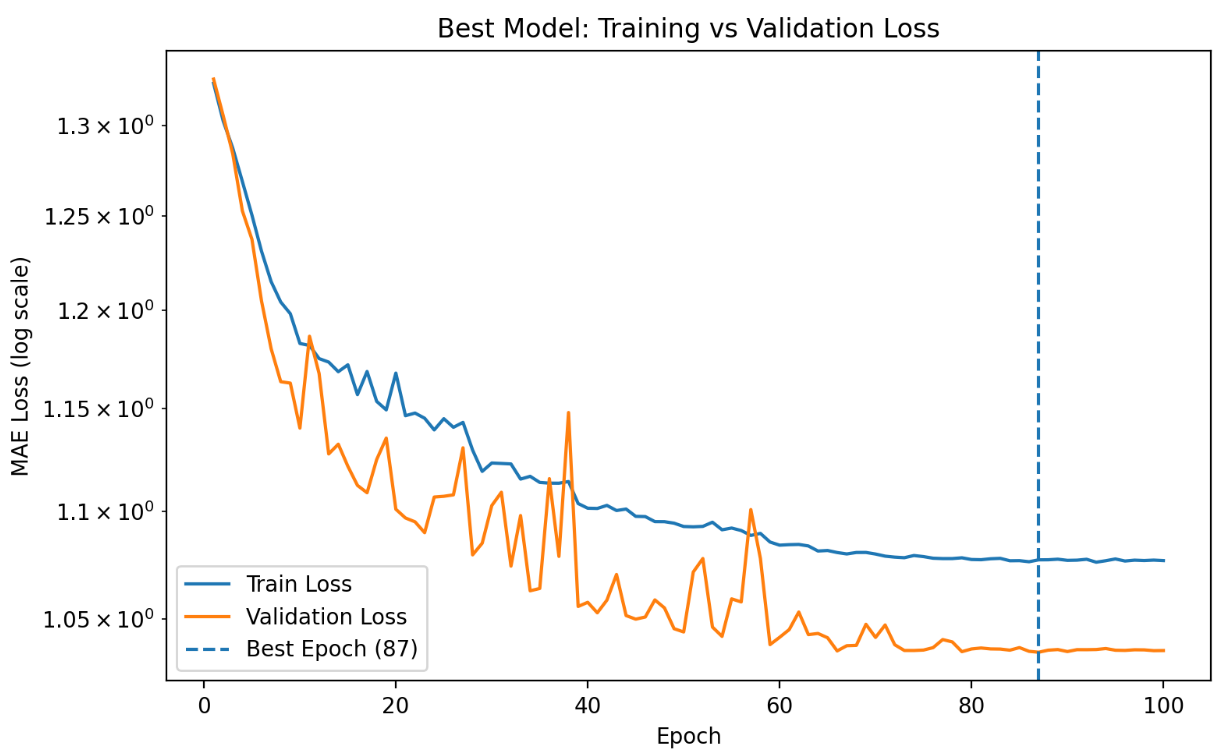

Figure 1: Best Model Loss Curves

Prediction quality of this model can be distinguished by the residual histogram, showing a roughly centered distribution near 0 with most residuals falling within -0.5 to +0.5 range. The rough bell curve confirms that the model is making strong predictions without systematic bias, with the heavier tails beyond +/-1 being caused by the harder-to-predict atoms at the charge extremes. This GNN model's behavior can be determined by the residual vs. true charge plots, which reveals a systematic fan-shaped pattern within each charge cluster where the prediction errors grow proportionally with the magnitude of the true charge. This is a known characteristic of all models trained with Mean Square Error loss, indicating that while the model has learned a strong average representation of how the charge is distributed, its power to discriminate is still limited at the extreme charge values.

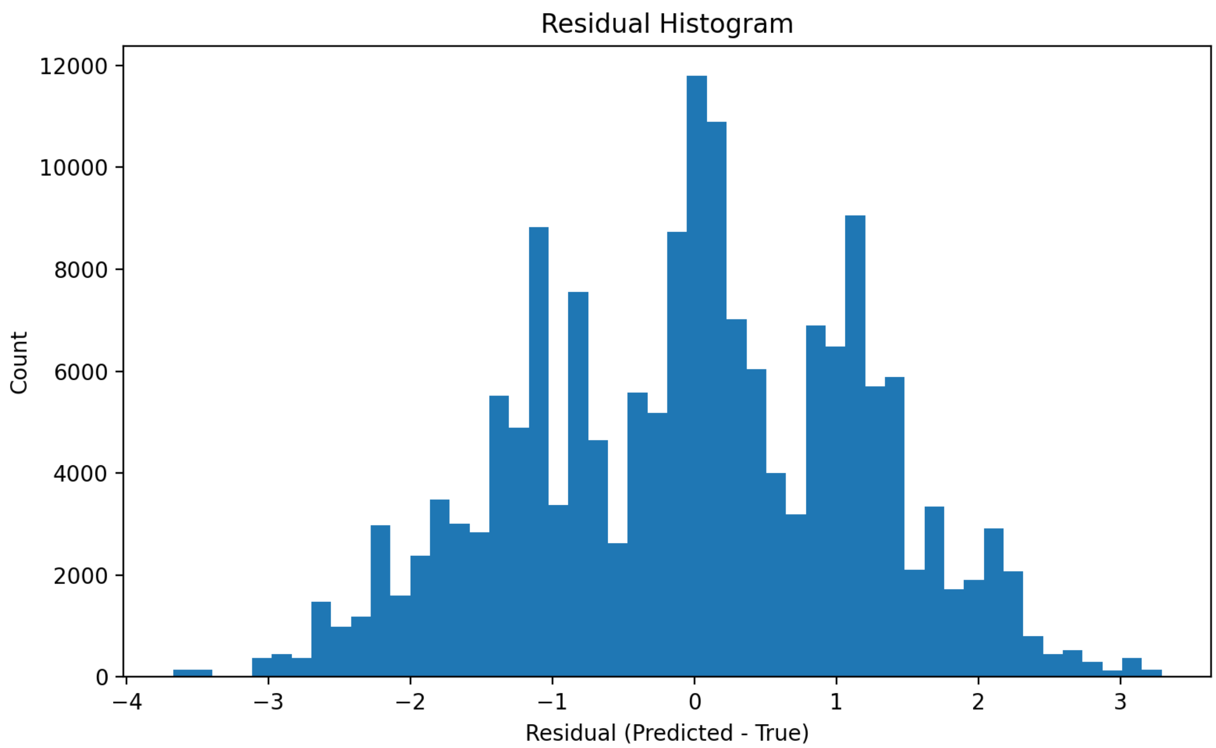

Figure 2 – Residual Histogram

The ability for the model to capture the overall charge distribution across the dataset can be seen through the predicted vs. true atomic charge scatterplot, where it has formed two dense clusters representing the two dominant charge groups in the dataset near -1.2 and +1.3. Both charge groups roughly align with the ideal diagonal lines, indicating that the model can correctly identify the charge sign and the approximate magnitude for most atoms. On the other hand, a slight compression effect is visible at the two different charge extremes with predictions for atoms with true charges beyond +/-1.5 tend to be pulled toward the center. This suggests that the model is slightly underpredicting at the tails of the charge distribution.

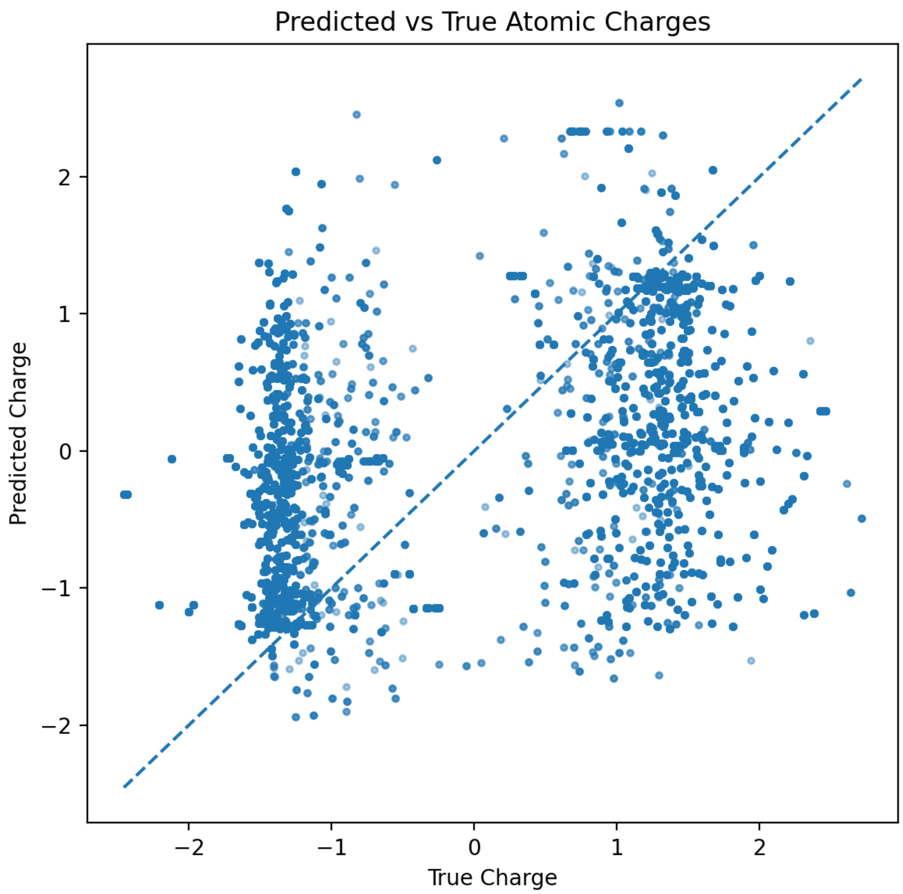

Figure 3 – Prediction vs. True Atomic Charges

For a comprehensive model evaluation, a sample of 172208 atomic charges were fed into the model for prediction. The results have demonstrated strong and accurate predictive performance. With a R^2 score of 0.959, this has confirmed that the model was able to capture around 96% of atomic charge variance across the test set with only a small fraction of the data left unexplained. Given the diversity and complexity of the POSCAR and atomic charge files, the accuracy that this model achieved is exceptionally high. The prediction error of this model is also important to analyze, as it can provide insights into where potential improvements can be made to the model to further increase its accuracy. Throughout the entire training epochs, this model has achieved a Mean Absolute Error (MAE) of 0.197 and a Root Mean Square Error (RMSE) of 0.274. The fact that the RMSE is greater than the MAE directly correlates to the harder-to-predict atoms at charge extremes. However, the low Mean Square Error (MSE) of 0.075 confirms that the model is unaffected by the large sporadic errors, signifying that its strong performance is consistent rather than selective. Overall, these metrics have shown a well-trained model that has learned the underlying relationship between crystal structure and atomic charge distribution, outperforming the traditional quantum mechanical approach for this task.


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Figure 4 – Best training Epoch Output Stats

Discussion & Future Work

This model has successfully been trained to predict atomic charges from crystal structure based on the quantitative and qualitative results from the GNN model. The high R^2 score of 0.959, along with the training and validation loss curve, indicates that the model was able to capture most of the atomic charge variance from the testing dataset. The consistent convergence throughout the training with a small gap between the validation and training loss curve confirms the model's ability to generalize learned features instead of memorizing them individually.

The residual analysis, however, has revealed some limitations of this model when it comes to predicting atomic charges for atoms on the two extremes of the charge distribution. One of the known limitations with all models trained with MSE loss was observed through the fan-shaped patterns in the residual vs. true charge plots where the prediction errors grow proportionally with the magnitude of the true charge. Regression towards the mean effect has shown signs that even though the model has learned a strong average representation of the charge distribution, it still struggles to fully discriminate between atoms with more extreme charge values. In addition, the compression effect from the predicted vs. true charge scatterplot further supports this effect where predictions for atoms beyond +/- 1.5 tend to pull toward the cluster center, rather than reaching the true extreme values.

The two-stage hyperparameter search has been an effective strategy when identifying a well-performing model configuration. This graph ML model's performance was highly dependent on the hyperparameter choices, as observed from the wide range of validation losses across the 20 epochs from the Stage-1 trial. Therefore, a systematic hyperparameter search was necessary rather than relying on manual tuning alone. Through multiple rounds of testing the different combinations, the best performing configuration has been determined. This combination consisted of 3 convolutional layers, each with 64 neurons. The training process includes a 10% dropout rate and incorporated the AdamW optimizer with ELU activation function for fine tuning the data, resulting in an effective balance between model complexity and generalization

Future improvements to this model can be made to enhance the quality of the model's predictive skills, including incorporating more expressive GNN architecture that encodes three-dimensional geometric information explicitly. This can help the model to capture the spatial relationships between atoms that influence charge distribution. Furthermore, meaningful chemical properties like electronegativity, atomic radius and coordination number can be used to enrich the node features; providing the model with more context about the local environment for each atom. To help address the regression toward the mean effect observed from the model, a weighted loss function specifically targets errors at the charge extremes can be integrated into the model. Lastly, a more diverse crystal structure dataset could be used to train the model, allowing it to generalize a wider range of material systems.